Effat College of Engineering · Computer Science

CS3081 Artificial Intelligence
Assignment: Uncertainty and Bayesian Networks

Instructor: Dr. Naila Marir **Total:** 50 marks (5 exercises × 10 marks)

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Instructions

1. Show all your steps. If you write only the final number, you get at most half of the marks.
2. Write probabilities with 4 decimal places (for example 0.5367), unless the question says something else.
3. Exercises 1 to 4, and Exercise 5 parts (a) and (b), are solved by hand.
4. Exercise 5 part (c) is solved in Python with pgmpy in VS Code, like in the lab.
5. Submit two files on Blackboard: these pages with your answers written in the boxes, scanned as one PDF, and your notebook `assignment_bn.ipynb` with all outputs.
6. This is individual work. You can discuss ideas with classmates, but you must write your own answers.

Exercise 1 — Joint distribution (10 marks)

The Effat shuttle bus is either **on time** or **late**. The weather in Jeddah is either **clear** or **dusty**. The table gives the probability of each pair. For example, the probability that the weather is clear **and** the bus is late is 0.14.

Weather	Shuttle = on time	Shuttle = late
clear	0.56	0.14
dusty	0.12	0.18

1. **(2 marks)** Show that the four numbers in the table make a valid probability distribution.

   All four numbers are between 0 and 1 (no negative numbers, none above 1). Sum: 0.56 + 0.14 + 0.12 + 0.18 = 1.0000. So the table is a valid probability distribution

2. **(2 marks)** Find P(dusty), the probability that the weather is dusty. Then find P(late), the probability that the bus is late.

   P(dusty) = P(dusty, on time) + P(dusty, late) = 0.12 + 0.18 = 0.3000
   P(late) = P(clear, late) + P(dusty, late) = 0.14 + 0.18 = 0.3200

3. **(2 marks)** Find P(late | dusty) and P(late | clear).

P(late | dusty) = P(dusty, late) / P(dusty) = 0.18 / 0.30 = 0.6000
P(late | clear) = P(clear, late) / P(clear) = 0.14 / (0.56 + 0.14) = 0.14 / 0.70 = 0.2000

4. **(2 marks)** Find P(dusty | late). Is it the same as P(late | dusty)? Explain your answer in one sentence.

P(dusty | late) = P(dusty, late) / P(late) = 0.18 / 0.32 = 0.5625
It is not the same as P(late | dusty) = 0.6000: the two conditional probabilities divide by different totals (P(late) = 0.32 and P(dusty) = 0.30).

5. **(2 marks)** Are the weather and the bus independent? Use numbers to explain your answer.

if they were independent, P(clear) × P(late) = 0.70 × 0.32 = 0.2240
would equal P(clear, late) = 0.14. But 0.2240 ≠ 0.14, so the weather and the bus are not independent. Also: P(late | dusty) = 0.6000 ≠ P(late) = 0.3200 (dust makes the bus more likely to be late).

Exercise 2 — Bayes' rule: a spam filter (10 marks)

An email program wants to find spam emails. It knows these facts:

- 30% of all emails are spam. So P(spam) = 0.3 and P(normal) = 0.7.
- In spam emails, the word "free" appears 60% of the time: P(free | spam) = 0.6.
- In normal emails, the word "free" appears 5% of the time: P(free | normal) = 0.05.

1. **(3 marks)** Find P(free), the probability that any email contains the word "free". Write the name of the rule you use.

Rule used: Total Probability (sum over spam / normal).
P(free) = P(free | spam) P(spam) + P(free | normal) P(normal)
= 0.6 × 0.3 + 0.05 × 0.7 = 0.18 + 0.035 = 0.2150

2. **(3 marks)** A new email contains the word "free". Use Bayes' rule to find P(spam | free).

Bayes' rule: P(spam | free) = P(free | spam) P(spam) / P(free)
= (0.6 × 0.3) / 0.2150 = 0.18 / 0.2150 = 0.8372

3. **(3 marks)** The program also checks the word "winner": P(winner | spam) = 0.4 and P(winner | normal) = 0.02. Assume that, once we know if an email is spam or normal, the two words appear independently of each other. A new email contains **both** "free" and "winner". Find P(spam | free, winner).

Words are independent given the class, so:
spam: P(spam) P(free | spam) P(winner | spam) = 0.3 × 0.6 × 0.4 = 0.0720
normal: P(normal) P(free | normal) P(winner | normal) = 0.7 × 0.05 × 0.02 = 0.0007 Normalization: 0.0720 + 0.0007 = 0.0727 P(spam | free, winner) = 0.0720 / 0.0727 = 0.9904

4. **(1 mark)** Compare your answers to parts 2 and 3. Did the probability of spam go up or down when we added the word "winner"? Explain why in one sentence.

t went up (0.8372 → 0.9904), because "winner" is much more likely in spam (0.4) than in normal email (0.02), so it is strong extra evidence for spam.

Exercise 3 — Bayesian network: structure and joint probability (10 marks)

In Jeddah summers, a **heatwave** (H) can cause an **AC overload** (A) in a campus building. A heatwave and an AC overload can both cause a **power cut** (C). A power cut can cause the **lab session to be cancelled** (L). Each variable is yes or no.

The network has 4 arrows: H → A, H → C, A → C, and C → L. The tables give only the probability of "yes". The probability of "no" is 1 minus the probability of "yes".

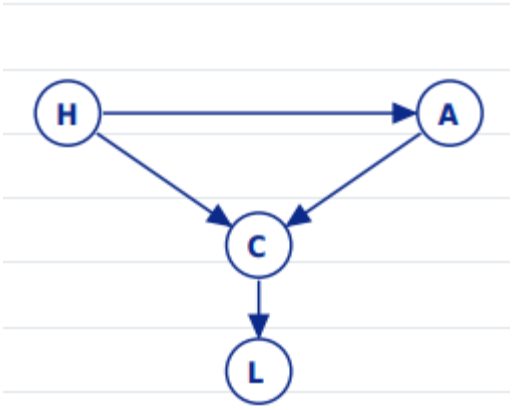
P(H = yes)
0.3

H	P(A = yes | H)
yes	0.6
no	0.1

H	A	P(C = yes | H, A)
yes	yes	0.7
yes	no	0.2
no	yes	0.5
no	no	0.05

C	P(L = yes | C)
yes	0.8
no	0.1

1. **(2 marks)** Draw the network. For each node, write its parents.



Parents of each node:
H: none (no parents)
A: H
C: H, A
L: C
Arrows: H → A, H → C, A → C, C → L

2. **(2 marks)** Write P(H, A, C, L) as a product of the probabilities from the network's tables.

P(H, A, C, L) = **P(H) · P(A | H) · P(C | H, A) · P(L | C)**

3. **(2 marks)** A full joint table for 4 yes/no variables needs 15 numbers ($2^4 - 1$). How many numbers do the 4 tables of this network need? Count only the "yes" probabilities, and show how you counted.

H needs 1 number (P(H=yes)).
A needs 2 numbers (one for each value of H).
C needs 4 numbers (one for each pair (H, A)).
L needs 2 numbers (one for each value of C).
Total: 1 + 2 + 4 + 2 = **9 numbers**, compared with 15 for the full joint table.

4. **(2 marks)** Find P(H = yes, A = no, C = yes, L = yes).

P(H=yes, A=no, C=yes, L=yes) = P(H=yes) · P(A=no | H=yes) · P(C=yes | H=yes, A=no) · P(L=yes | C=yes)
= 0.3 × 0.4 × 0.2 × 0.8 = **0.0192**

5. **(2 marks)** Find P(H = yes | C = yes). Then give P(H = yes | C = yes, L = yes) without doing a new calculation, and explain why the two answers are the same.

P(H=yes, C=yes) = 0.3 × (0.6×0.7 + 0.4×0.2) = 0.3 × 0.50 = 0.1500
P(H=no, C=yes) = 0.7 × (0.1×0.5 + 0.9×0.05) = 0.7 × 0.095 = 0.0665; P(C=yes) = 0.2165
P(H=yes | C=yes) = 0.1500 / 0.2165 = **0.6928**
P(H=yes | C=yes, L=yes) = **0.6928** (same). C is the only parent of L, so given C, L is independent of H (L ⊥ H | C): learning L adds no information about H.

Exercise 4 — Inference by enumeration, challenge question (10 marks)

This is the **challenge question** of the assignment. Show every step.

Use the network and tables from Exercise 3. Today the lab session **was cancelled** (L = yes).

1. **(4 marks)** Use inference by enumeration to find P(H = yes | L = yes), the probability that there was a heatwave. Write the query variable, the evidence and the hidden variables, then show all sums and the normalization.

1) Query variable: **H**. Evidence: **L = yes**. Hidden variables: **A and C**.
$P(H, L=yes) = \Sigma_A \Sigma_C P(H) P(A|H) P(C|H,A) P(L=yes|C)$

H = yes:

A	C	Product	Value
yes	yes	0.3 × 0.6 × 0.7 × 0.8	0.1008
yes	no	0.3 × 0.6 × 0.3 × 0.1	0.0054
no	yes	0.3 × 0.4 × 0.2 × 0.8	0.0192
no	no	0.3 × 0.4 × 0.8 × 0.1	0.0096
		Sum	**0.1350**

H = no:

A	C	Product	Value
yes	yes	0.7 × 0.1 × 0.5 × 0.8	0.0280
yes	no	0.7 × 0.1 × 0.5 × 0.1	0.0035
no	yes	0.7 × 0.9 × 0.05 × 0.8	0.0252
no	no	0.7 × 0.9 × 0.95 × 0.1	0.05985
		Sum	**0.11655**

Normalization: 0.1350 + 0.11655 = 0.25155
P(H=yes | L=yes) = 0.1350 / 0.25155 = **0.5367** (P(H=no | L=yes) = 0.4633)

2. **(3 marks)** Now find P(A = yes | L = yes), the probability that there was an AC overload. Which variables are hidden this time?

2) Query: A. Evidence: L = yes. Hidden variables: **H and C**.
A = yes: (0.1008 + 0.0054) + (0.0280 + 0.0035) = 0.1062 + 0.0315 = 0.1377
A = no: (0.0192 + 0.0096) + (0.0252 + 0.05985) = 0.0288 + 0.08505 = 0.11385
Normalization: 0.1377 + 0.11385 = 0.25155
P(A=yes | L=yes) = 0.1377 / 0.25155 = **0.5474**

3. **(3 marks)** A technician checks the building and reports that there was **no** AC overload (A = no). The lab is still cancelled (L = yes). Find P(H = yes | L = yes, A = no). Compare it with your answer to part 1: did the probability of a heatwave go up or down? Use the network to explain why in two or three sentences.

> Query: H. Evidence: L = yes, A = no. Hidden: C.
> H = yes: 0.3 × 0.4 × [P(C=yes|H=yes,A=no) × 0.8 + P(C=no|H=yes,A=no) × 0.1] = 0.12 × [0.2×0.8 +
> 0.8×0.1] = 0.0192 + 0.0096 = 0.0288
> H = no: 0.7 × 0.9 × [0.05×0.8 + 0.95×0.1] = 0.0252 + 0.05985 = 0.08505
> Normalization: 0.0288 + 0.08505 = 0.11385
> P(H=yes | L=yes, A=no) = 0.0288 / 0.11385 = 0.2530
> Comparison: it went down (0.5367 → 0.2530). A heatwave makes an AC overload much more likely (0.6
> vs 0.1), so learning that there was NO overload is evidence against a heatwave. The cancelled lab is then
> better explained by a power cut that happened without a heatwave (H → C directly, or by chance).

Exercise 5 — Approximate inference by sampling (10 marks)

Now we find the same probability, P(H = yes | L = yes), by **sampling**. Sampling gives an estimate, not the exact answer.

(a) Rejection sampling (3 marks). Reminder: in rejection sampling, we keep only the samples that agree with the evidence (here L = yes), and we throw away the others. Below are 12 samples from the network of Exercise 3.

#	H	A	C	L
1	no	no	no	no
2	yes	yes	yes	yes
3	no	no	no	no
4	no	yes	yes	yes
5	yes	yes	no	no
6	no	no	no	yes
7	yes	no	yes	yes
8	no	no	no	no
9	no	no	no	no
10	yes	yes	yes	no
11	no	no	no	no
12	no	no	yes	yes

1. Write the numbers of the samples we keep and the samples we throw away. Then estimate P(H = yes | L = yes).

Keep samples with L = yes: 2, 4, 6, 7, 12. Throw away: 1, 3, 5, 8, 9, 10, 11.
Among the kept samples, H = yes in samples 2 and 7. Estimate: P(H=yes | L=yes) ≈ 2/5 =
0.4000.

2. What fraction of the 12 samples did we throw away? Why is this a problem for rejection sampling?

Thrown away: 7/12 = 0.5833 (about 58%). This is a problem because most of the work is
wasted. If the evidence is rare, almost all samples are rejected and we need a huge number of
samples to get a good estimate.

(b) Likelihood weighting (3 marks). Reminder: in likelihood weighting, we do not sample the evidence. We fix it (here L = yes), we sample the other variables, and we give each sample a **weight**. The weight is the probability of the evidence given the sample's values: here, P(L = yes | C). Below are 6 samples.

#	H	A	C	L
1	no	no	no	yes (fixed)
2	yes	yes	yes	yes (fixed)
3	no	no	no	yes (fixed)
4	yes	no	no	yes (fixed)
5	no	yes	yes	yes (fixed)
6	yes	yes	yes	yes (fixed)

1. Write the weight of each sample.

Weight = P(L=yes | C): it is 0.8 if C = yes, and 0.1 if C = no.
Sample 1 (C=no): 0.1 Sample 2 (C=yes): 0.8 Sample 3 (C=no): 0.1
Sample 4 (C=no): 0.1 Sample 5 (C=yes): 0.8 Sample 6 (C=yes): 0.8

2. Estimate P(H = yes | L = yes) using the weights: add the weights of the samples where H = yes, then divide by the sum of all weights.

Weights where H = yes (samples 2, 4, 6): 0.8 + 0.1 + 0.8 = 1.7. Sum of all weights:
0.1+0.8+0.1+0.1+0.8+0.8 = 2.7.
P(H=yes | L=yes) ≈ 1.7 / 2.7 = 0.6296

3. Compare your estimates from (a) and (b) with the exact answer from Exercise 4. Why are they different? How can we make them closer to the exact answer?

The exact answer is 0.5367, while (a) gives 0.4000 and (b) gives 0.6296. They differ because
sampling estimates are random and here are based on very few samples (5 useful samples in (a),
6 in (b)). To get closer to the exact answer, increase the number of samples. Likelihood weighting also uses every sample, so it wastes less than rejection sampling.

(c) Python with pgmpy (4 marks). Answer this part in your notebook, not on paper. In VS Code, create `assignment_bn.ipynb`.

1. Build the network of Exercise 3 with `DiscreteBayesianNetwork` and `TabularCPD`, like in the lab. Name the nodes `heatwave`, `overload`, `powercut` and `cancel`, and use the values `yes` and `no`. Print `model.check_model()`. It must print `True`.
2. Use `VariableElimination` to find P(H | L = yes). Your answer must be the same as in Exercise 4.
3. Use rejection sampling with 10,000 samples to estimate P(H = yes | L = yes). Hint: import `BayesianModelSampling` from `pgmpy.sampling` and `State` from `pgmpy.factors.discrete`. Then use `BayesianModelSampling(model).rejection_sample(evidence=[State("cancel", "yes")], size=10000)`. This gives a table of samples. Count the fraction of rows where `heatwave` is `yes`.
4. In a Markdown cell, write two sentences that compare the exact answer and the sampling estimate.